Module 5 — Factors, divisibility and mathematical loops

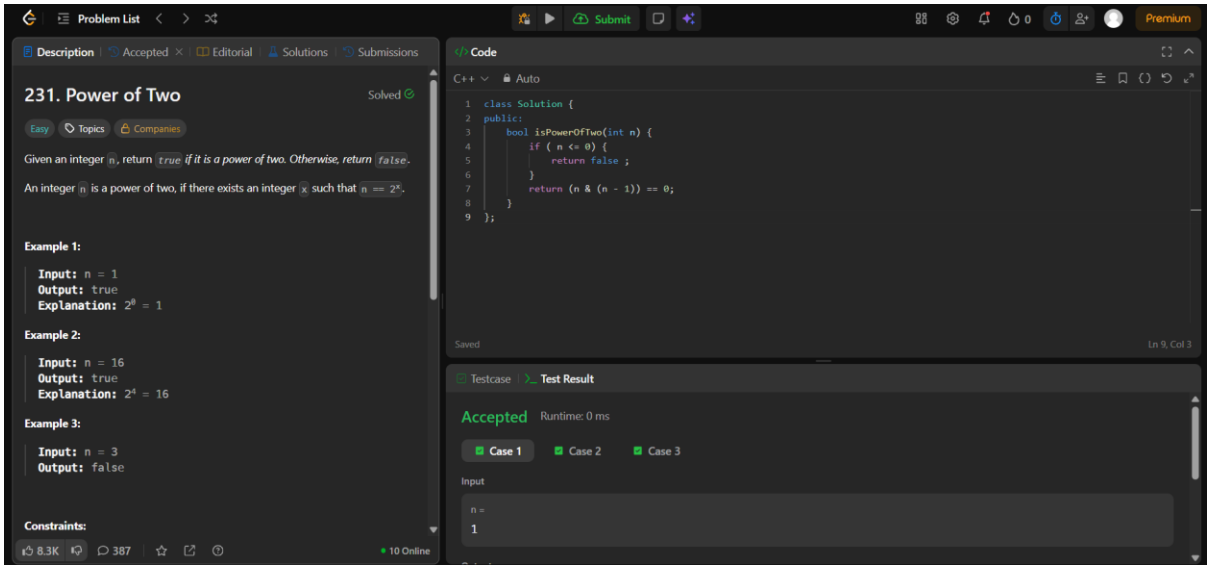
By Tanishka Gahlot

Questions

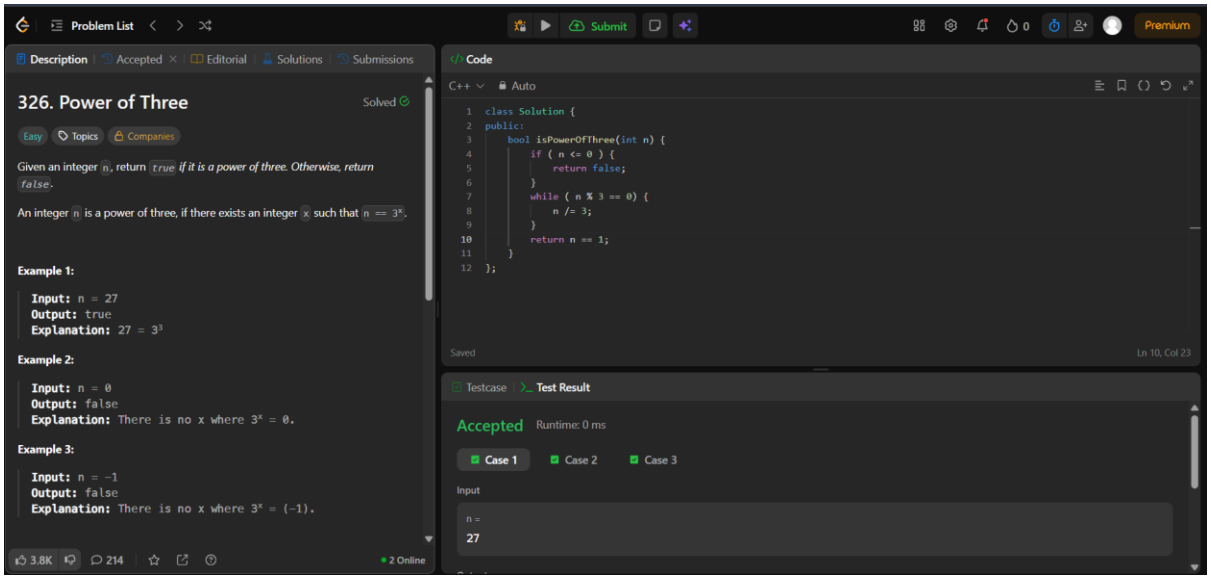
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No.	Problem (click to open)	Platform	Difficulty
1	231 — Power of Two	LeetCode	Easy
2	326 — Power of Three	LeetCode	Easy
3	Factorial!	HackerEarth	Easy
4	Count Divisors	HackerEarth	Easy–Moderate
5	Prime Numbers	HackerEarth	Easy–Moderate
6	507 — Perfect Number	LeetCode	Easy–Moderate

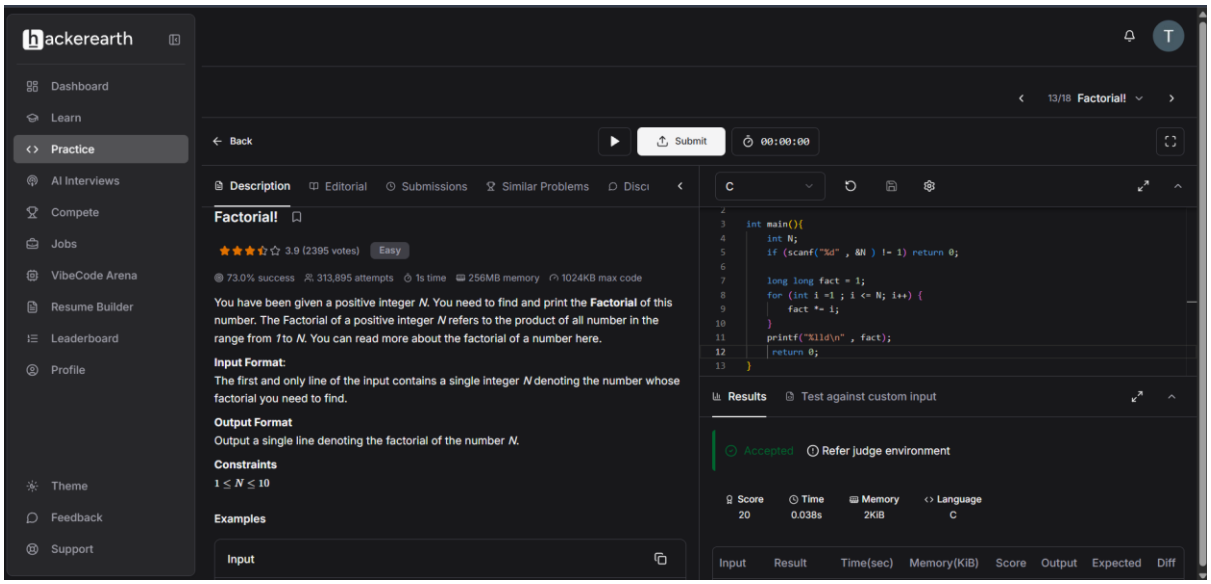
Question 1 = https://leetcode.com/problems/power-of-two/description/



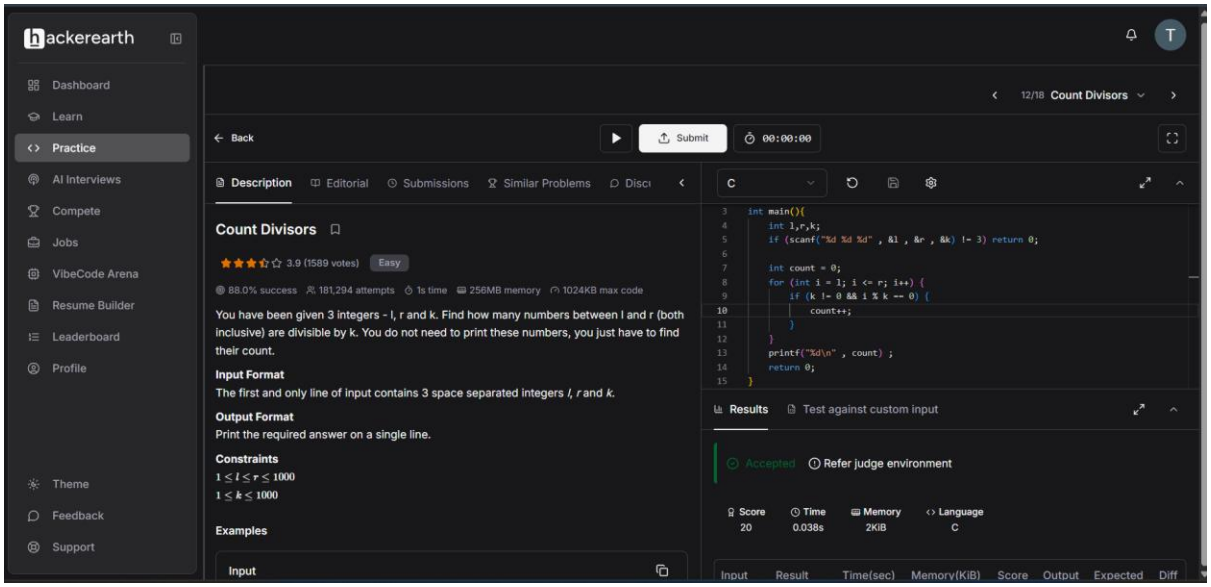
Question 2 = https://leetcode.com/problems/power-of-three/description/



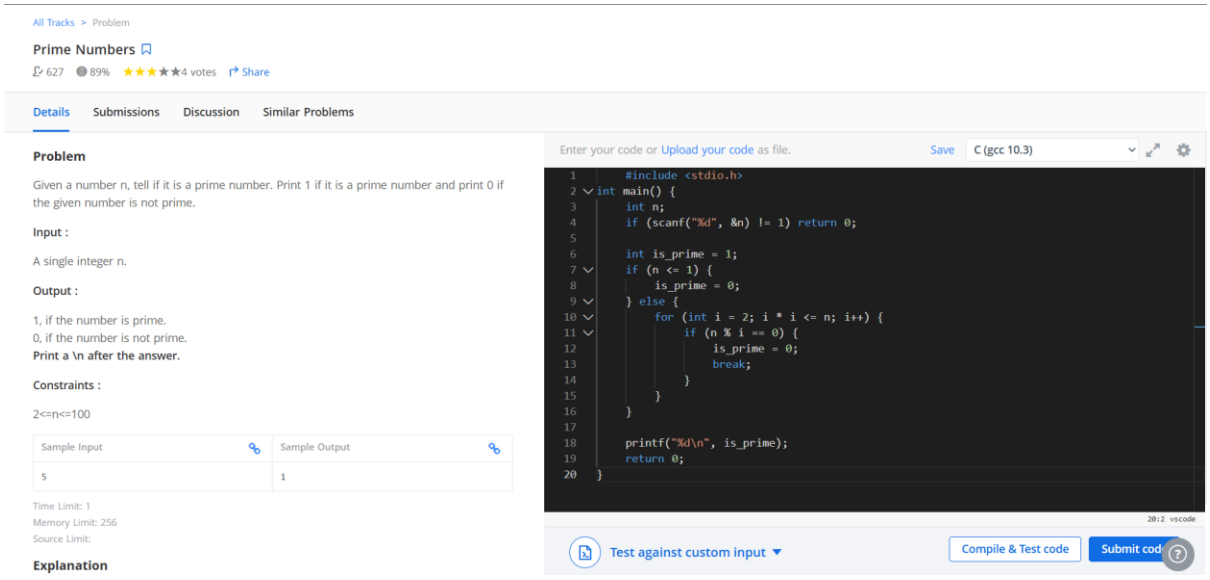
Question 3 = https://www.hackerearth.com/practice/basic-programming/input-output/basics-of-input-output/practice-problems/algorithm/find-factorial/

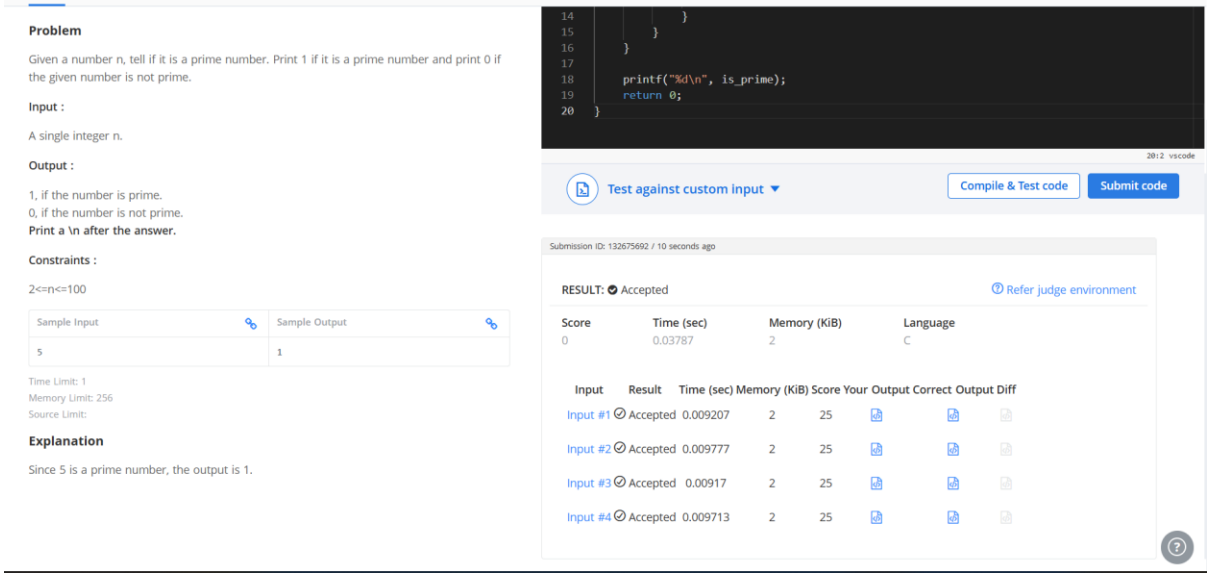


Question 4 = https://www.hackerearth.com/practice/basic-programming/input-output/basics-of-input-output/practice-problems/algorithm/count-divisors/

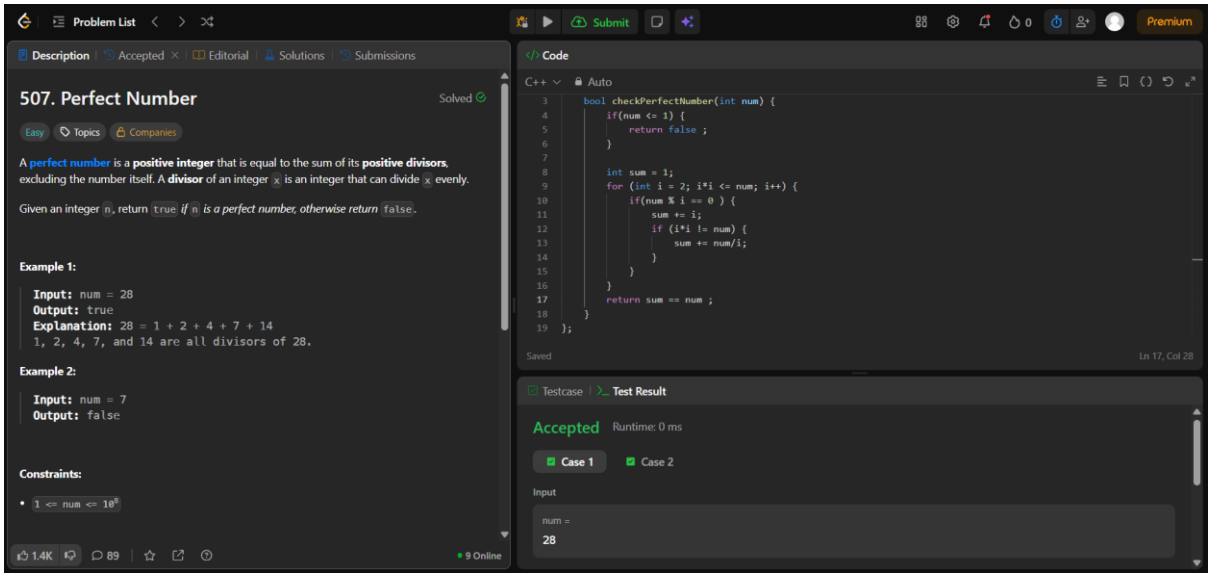


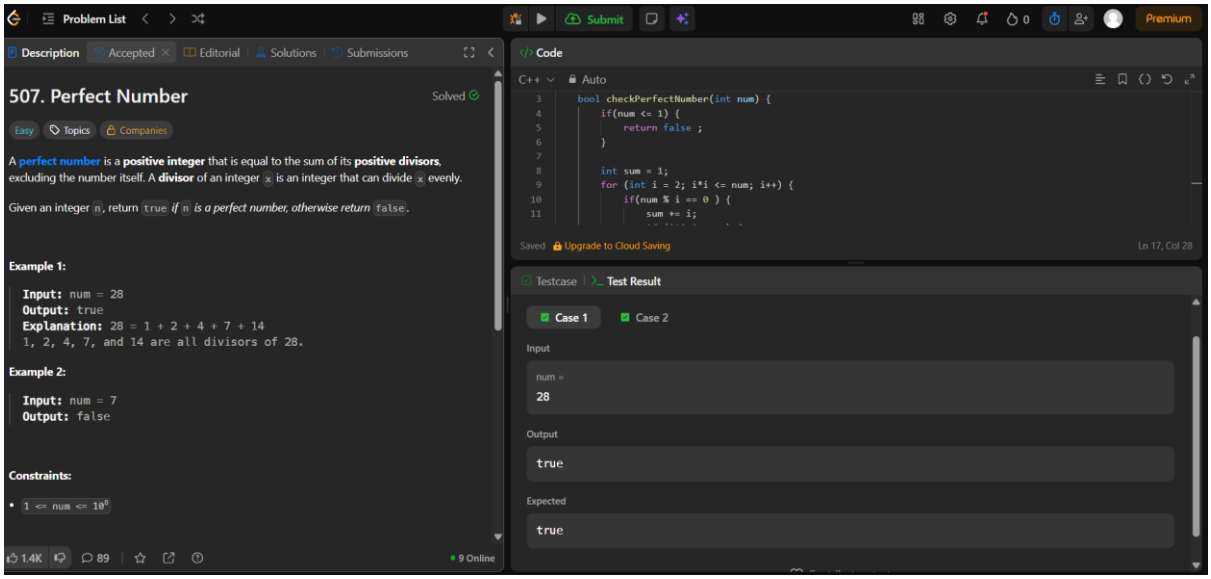
Question 5 = https://www.hackerearth.com/problem/algorithm/prime-numbers/





Question 6 = https://leetcode.com/problems/perfect-number/description/


Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
507. Perfect Number
Solved
Easy
Topics
Companies
A perfect number is a positive integer that is equal to the sum of its positive divisors, excluding the number itself. A divisor of an integer x is an integer that can divide x evenly.
Given an integer n, return true if n is a perfect number, otherwise return false.
Example 1:
Input: num = 28
Output: true
Explanation: 28 = 1 + 2 + 4 + 7 + 14
1, 2, 4, 7, and 14 are all divisors of 28.
Example 2:
Input: num = 7
Output: false
Constraints:
1 <= num <= 10^8
1.4K
89
9 Online
Code
C++
Auto
bool checkPerfectNumber(int num) {
if(num <= 1) {
return false ;
}
int sum = 1;
for (int i = 2; i*i <= num; i++) {
if(num % i == 0) {
sum += i;
if (i*i != num) {
sum += num/i;
}
}
}
return sum == num ;
}
};
Saved
Ln 17, Col 28
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Input
num =
28


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Saved
Upgrade to Cloud Saving
Ln 17, Col 28
Testcase
Test Result
Case 1
Case 2
Input
num =
28
Output
true
Expected
true